

DBMS Assignment 2 — Normalization & Schema Refinement

Database Management Systems (DBMS) · Unit 3 · Normalization & Schema Refinement · 2025

Answer all questions. Show working and cite references where used.

Questions

1. Discuss the role of Functional Dependencies in Normalization & Schema Refinement.
2. Explain Closure of Attributes with a neat diagram and a suitable example.
3. Illustrate First, Second and Third Normal Form with a worked example.
4. Discuss the role of Boyce-Codd Normal Form in Normalization & Schema Refinement.
5. Explain Lossless Join Decomposition with a neat diagram and a suitable example.
6. What is meant by Multivalued Dependencies and 4NF? Discuss its advantages and limitations.